

2023 Post-lab Quiz Answer Name:

Supporting Information for

Title:

Discovery-based approach to identify multiple factors that affect the spin state of coordination complexes using the Evans NMR method

Authors:

Masaki Uchida*, Candice H. Cortney, Karina Bustos, Elia Manzo, Emily Sauls, Jacob Bouchard, Risako Fukazawa, and Viswanathan V. Krishnan

Affiliations:

Department of Chemistry and Biochemistry, California State University, Fresno
2555 E. San Ramon Ave., MS SB70, Fresno, CA 93740, USA

*** Corresponding author: Masaki Uchida**

Email: muchida@mail.fresnostate.edu

1. What is the d electron counts of the following metal ions?

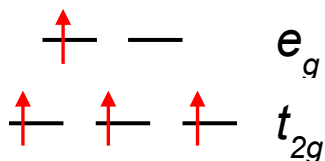
(Question Category 1 (QC 1), that corresponds to Table 2 and Figure 2)

(1) Cr^{3+} d^3 (2) Co^{2+} d^7 **2. Each of the following ions could form a coordination complex with octahedral geometry. Draw d-orbital splitting diagrams of both a) high-spin and b) low-spin octahedral complexes to show orbital occupancies with electrons.**

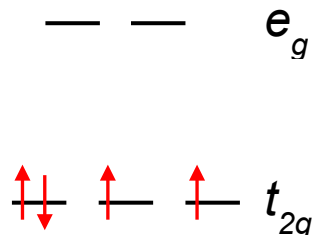
(QC 3)

(1) Mn^{3+}

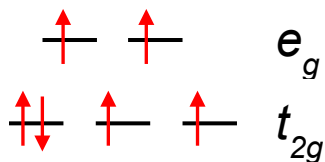
a) High-spin



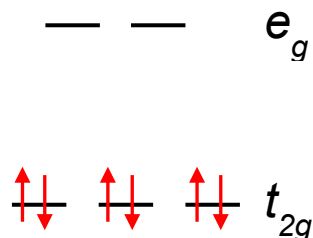
b) Low-spin

(2) Co^{3+}

a) High-spin



b) Low-spin



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3. For each of the following complexes, answer questions (a) – (c).

- (a) Determine the oxidation number of the metal. (QC 1)
(b) Determine the coordination number of the metal (i.e. the number of atoms or ions immediately surrounding the metal in the complex). (QC 2)
(c) What is the expected geometry of the complex. Select from the following i) – iv) (QC 2)
i) Trigonal planar, ii) Tetrahedral, iii) Square planar, iv) Octahedral

(1) $\text{Cr}(\text{acac})_3$

(a): Cr^{3+}

(b): 6

(c): iv

(2) $[\text{Fe}(\text{CN})_6]^{3-}$

(a): Fe^{3+}

(b): 6

(c): iv

(3) $[\text{FeCl}_4]^-$

(a): Fe^{3+}

(b): 4

(c): ii

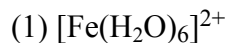
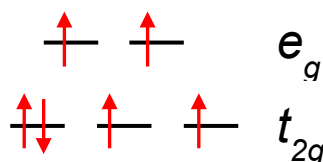
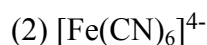
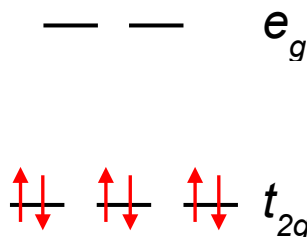
4. For each of the following complexes, answer questions (a) – (c).

(a) Predict whether this complex is high-spin or low-spin state. Briefly explain your reasoning.

(QC 4)

(b) What is the number of unpaired electrons in this complex? (QC 3)

(c) Is this complex paramagnetic or diamagnetic?

(a): **High-spin** H_2O is a relatively weak-field ligand. Most aqua complexes are high spin. An exception is $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, that is low-spin due to the central metal ion Co^{3+} .(b): **4**(c): **Paramagnetic**(a): **Low-spin** CN^- ligand acts as a π -acceptor. Thus it tends to show large Δ_o and is considered as a strong-field ligand.(b): **0**(c): **Diamagnetic**

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5. Answer the following questions.

(1) Which of the following complexes has the largest crystal field splitting energy Δ_o ? Briefly explain your reasoning. (QC 4)

(a): $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, (b): $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, (c): $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$

(c) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$

Δ_o increases with increasing oxidation state of the central metal ion.

The order of Δ_o is $\text{Co}^{2+} < \text{Fe}^{3+} < \text{Co}^{3+}$

(2) Which of the following complexes has the largest crystal field splitting energy Δ_o ? Briefly explain your reasoning. (QC 4)

(a): $\text{Fe}(\text{acac})_3$, (b): $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, (c): $[\text{Fe}(\text{CN})_6]^{3-}$,

(c): $[\text{Fe}(\text{CN})_6]^{3-}$

CN^- ligand is a strong-field ligand. H_2O and acac are relatively weak-field ligands.

(3) Which of the following parameters effect on the magnitude of the crystal field splitting energy Δ_o of coordination complexes? Select all. (QC 4)

(a): type of ligands

(b): type of metals

(c): oxidation state of metals

(d): concentration of coordination complexes in solution

Ans: (a) (b) (c)
